

STAT 17 Discussion Notes

Week 9: Hypothesis Testing

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1 Goal of Hypothesis Testing

Hypothesis testing is a statistical method used to determine whether there is enough evidence in sample data to support a claim about a population parameter.

Examples of questions answered using hypothesis tests:

- Does a new drug have fewer side effects?
- Has public opinion changed over time?
- Is the average body temperature different from 98.6°F?

A hypothesis test uses sample data to decide whether a population parameter is likely equal to a hypothesized value.

2 The Five Steps of a Hypothesis Test

1. State the Hypotheses

Two hypotheses are always defined:

- Null hypothesis (H_0): the status quo or assumed value
- Alternative hypothesis (H_1 or H_a): the claim we want evidence for

Three possible forms:

$$H_0 : \text{parameter} = \text{value}$$

Left-tailed:

$$H_1 : \text{parameter} < \text{value}$$

Right-tailed:

$$H_1 : \text{parameter} > \text{value}$$

Two-tailed:

$$H_1 : \text{parameter} \neq \text{value}$$

2. Check Conditions and Choose Significance Level

Common significance levels:

$$\alpha = 0.01, \quad 0.05, \quad 0.10$$

α represents the probability of making a Type I error.

3. Calculate the Test Statistic

The test statistic measures how far the sample statistic is from the hypothesized value in standard errors.

4. Find the p-value

The p-value is the probability of observing the sample result (or something more extreme) assuming H_0 is true.

5. Draw a Conclusion

- If $p\text{-value} \leq \alpha$: Reject H_0
- If $p\text{-value} > \alpha$: Do Not Reject H_0

Conclusions should always be written in terms of the **alternative hypothesis**.

3 Hypothesis Test for a Population Proportion

Used when the parameter is a population proportion p .

3.1 Conditions

- Random sample or randomized experiment
- Sample size less than 5% of population
- Large sample condition:

$$np_0(1 - p_0) \geq 10$$

3.2 Test Statistic

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

3.3 p-value

Left-tailed test:

$$p\text{-value} = P(Z < z)$$

Right-tailed test:

$$p\text{-value} = P(Z > z)$$

Two-tailed test:

$$p\text{-value} = 2P(Z > |z|)$$

4 Hypothesis Test for a Population Mean (Case 1: σ^2 is unknown)

Used when the parameter is a population mean μ , σ .

4.1 Conditions

- Random sample
- Sample size less than 5% of population
- Data from a normal population OR $n \geq 30$

4.2 Test Statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- Follows a t distribution
- Degrees of freedom: $df = n - 1$

4.3 p-value

Left-tailed:

$$p\text{-value} = P(t < t_0)$$

Right-tailed:

$$p\text{-value} = P(t > t_0)$$

Two-tailed:

$$p\text{-value} = 2P(t > |t_0|)$$

5 Hypothesis Test for a Population Mean (Case 2)

Used when the parameter is a population mean μ .

5.1 Conditions

- Random sample
- Sample size less than 5% of population
- Data from a normal population OR $n \geq 30$

5.2 Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

5.3 p-value

Left-tailed:

$$p\text{-value} = P(Z < z_0)$$

Right-tailed:

$$p\text{-value} = P(Z > z_0)$$

Two-tailed:

$$p\text{-value} = 2P(Z > |z_0|)$$

6 Relationship Between Confidence Intervals and Hypothesis Tests

For a two-sided test:

- If the hypothesized value is **outside** the confidence interval $\Rightarrow$ Reject H_0
- If the hypothesized value is **inside** the confidence interval $\Rightarrow$ Do Not Reject H_0

Example:

A 95% confidence interval that contains the null value means the hypothesis test at $\alpha = 0.05$ will not reject H_0 .

7 Type I and Type II Errors

Because hypothesis tests use sample data, mistakes are possible.

7.1 Type I Error

Rejecting H_0 when H_0 is actually true.

$$P(\text{Type I error}) = \alpha$$

Example: Concluding a drug has an effect when it actually does not.

7.2 Type II Error

Failing to reject H_0 when H_1 is actually true.

$$P(\text{Type II error}) = \beta$$

Example: Concluding a drug has no effect when it actually works.

7.3 Important Relationship

Reducing α increases β .

8 Key Summary

Hypothesis testing process

1. State H_0 and H_1
2. Check conditions
3. Compute test statistic
4. Find p-value
5. Compare with α and conclude